



From lecture session 3: We learned how plate tectonics help forming an ocean basin and learned about some of the ocean topographic features related to plates' boundaries (Ex: Mid-Atlantic Ridges and Trenches)

Lecture Session 4

1. How scientists map and study the seafloor features;
2. The continental margins;
3. The abyssal plain and deep ocean features.

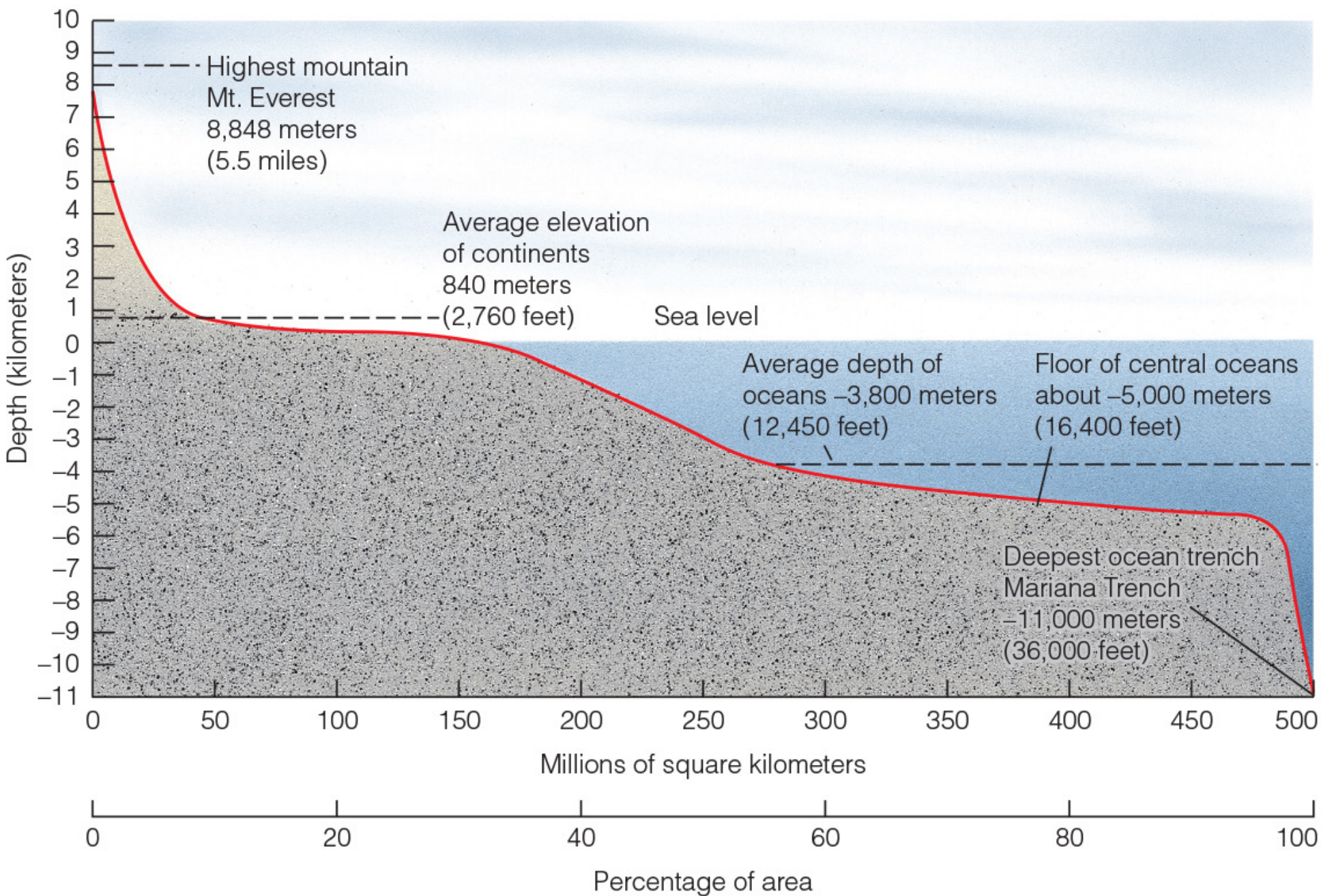


Figure 4.7 A graph showing the distribution of elevations and depths on Earth. This curve is not a land-to-sea profile of Earth, but rather a plot of the area of Earth's surface above any given elevation or depth below sea level. Note that more than half of Earth's solid surface is at least 3,000 meters (10,000 feet) below sea level. The average depth of the world ocean (3,796 meters or 12,451 feet) is much greater than the average height of the continents (840 meters or 2,760 feet)

Mapping the Ocean Floor

A challenging and expensive endeavor

Modern tools to map the ocean floor

1. Echo Sounders: “Cheap” for oceanographers’ standards; highly detailed mapping; and time consuming.

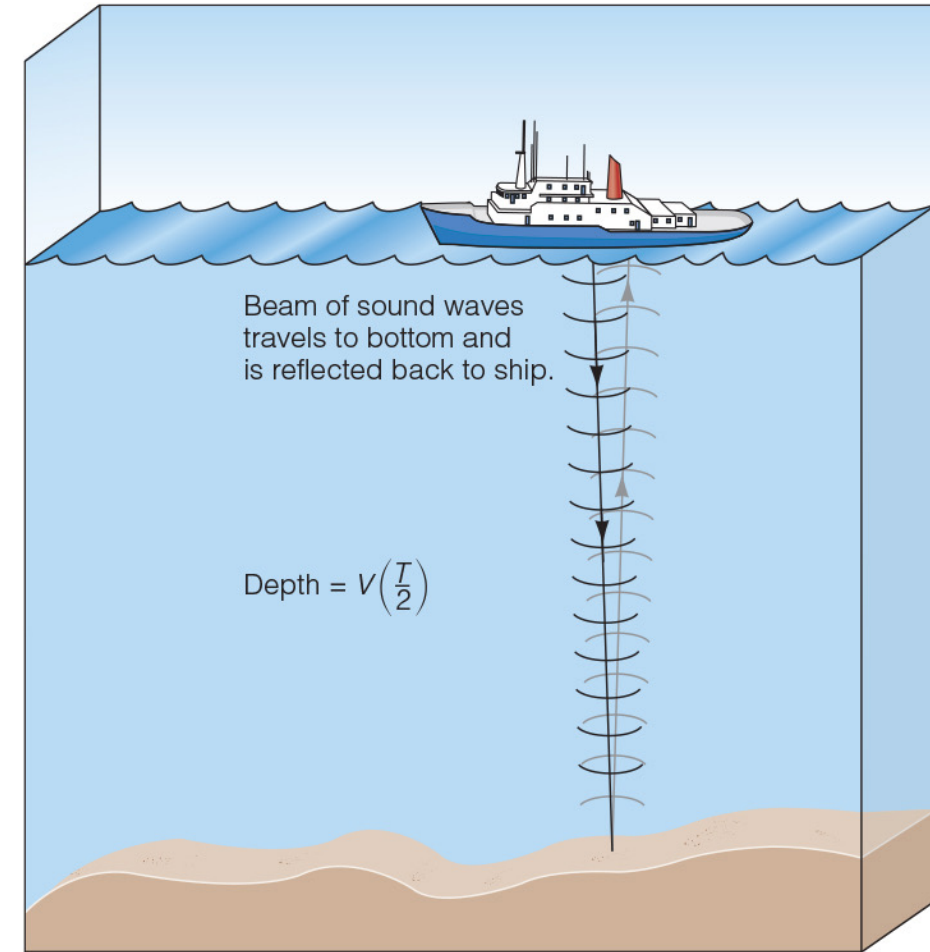


Figure 2.25 Echo sounders sense the seafloor’s contour by beaming sound waves to the bottom and measuring the time required for sound waves to bounce back to the ship. If the round-trip travel time and wave velocity are known, distance to the bottom can be calculated. This technique was first used on a large scale by the German research vessel *Meteor* in the 1920s.

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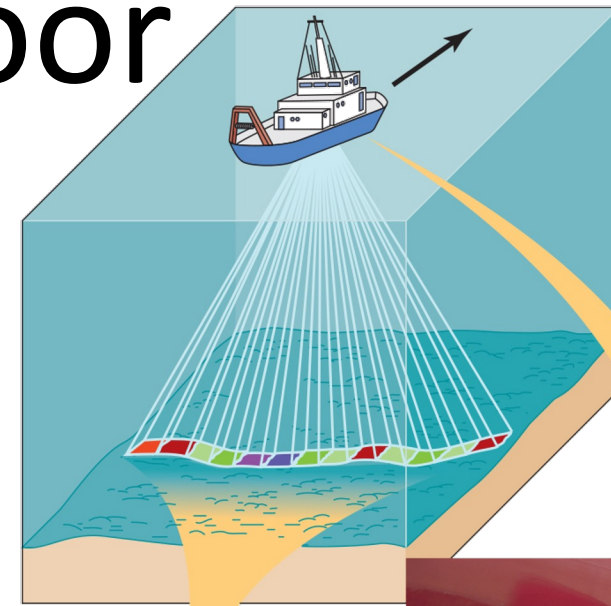
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1. Echo Sounders: “Cheap” for oceanographers’ standards; highly detailed mapping; and time consuming.

2. Multibeam Echo Sounders: Expensive and sophisticated system, high accuracy and level of mapping detail; better than simple single-pulse echo-sounders but still time consuming.

Both systems use sound. Why not electromagnetic waves (aka light)?

Sound propagate better underwater (low-attenuation)



a A multibeam echo sounder uses as many as 121 beams radiating from a ship's hull. Fanning out at right angles to the direction of travel, these beams can cover a 120° arc and measure a swath of bottom about 3.4 times as wide as the water is deep. Typically, a “ping” is sent toward the seabed every 10 seconds. Listening devices record sounds reflected from the bottom, but only from the narrow corridors corresponding to the outgoing pulse. Thus a multi-beam system is much less susceptible to error than the single-beam device shown in Figures 2.25 and 4.2.



b A multibeam transducer mounted beneath the bow of a research vessel.



Andrew Goodwillie/Scripps Institution of Oceanography, UC San Diego

c A multibeam record of a fragment of seafloor near the East Pacific Rise south of the tip of Baja California, Mexico. The uneven coverage reflects the path of the ship across the surface. Detailed analysis requires sailing a careful back-and-forth pattern. Computer processing provides extraordinarily detailed images, such as those in Figures 4.4e and 4.13.

Figure 4.3 How a multibeam system collects data.

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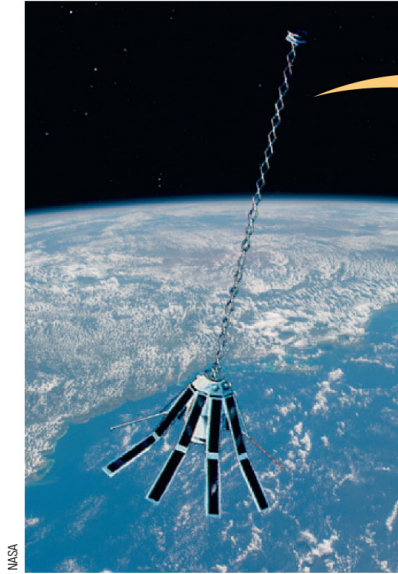
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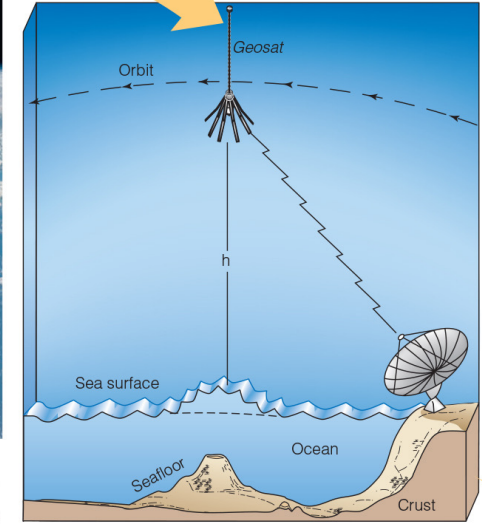
3. Satellites: Expensive to launch (we need a rocket), but once in orbit it can cover the entire planet in a few days. Satellites are not as precise as Multibeam Echo Sounders, however, large a prominent features can be studied in great detail.

Combining echo sounders mapping and satellite, scientists could produce incredible detailed mapping of the ocean topography

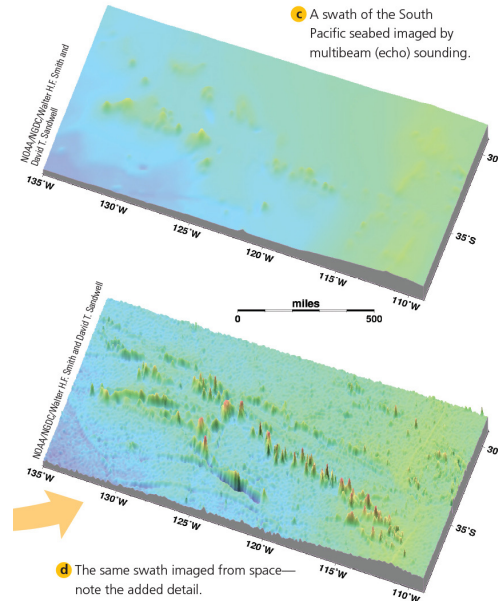
Figure 4.4 Measuring the seabed: a comparison of technologies.



a Geosat, a U.S. Navy satellite that operated from 1985 through 1990, provided measurements of sea surface height from orbit. Moving above the ocean surface at 7 kilometers (4 miles) a second, Geosat bounced 1,000 pulses of radar energy off the ocean surface every second. Height accuracy was within 0.03 meter (1 inch)! Multiple other satellites have since taken its place.



b Distortion of the sea surface above a seabed feature occurs when the extra gravitational attraction of the feature “pulls” water toward it from the sides, forming a mound of water over itself.



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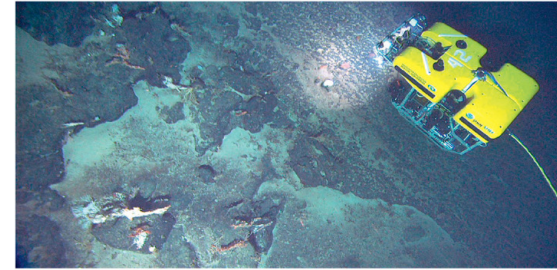
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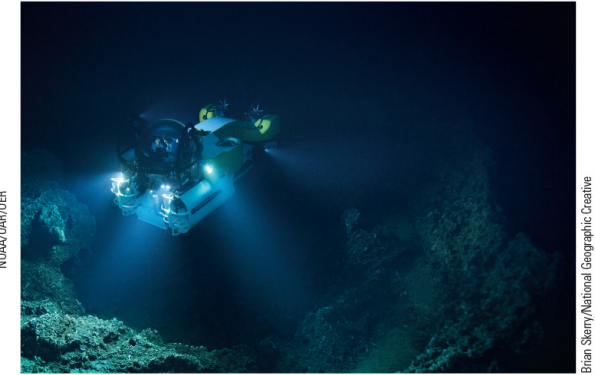
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4. Submersibles and bottom sampling:

Allow us visual inspection of the deep ocean and study its composition in detail. Because it requires ship time is expensive and time consuming.



a Operated by the Mystic Aquarium’s Institute for Exploration in conjunction with the University of Rhode Island and NOAA, the ROV *Hercules* is photographed from the ROV *Argo*. *Hercules* is tethered to *Argo* by a 30-meter (100-foot) cable. *Argo* was used to discover the remains of the RMS *Titanic* in 1985.



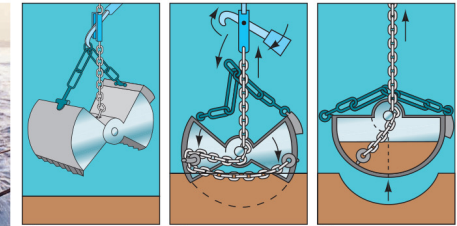
b The manned submersible, *DeepSee*, investigating a seamount as part of a National Geographic expedition to the Las Gemelas Seamounts near Cocos Island. Like the previous photo, the ROV *Argo* was also used to photograph this image.

Figure 4.5 Submersibles.



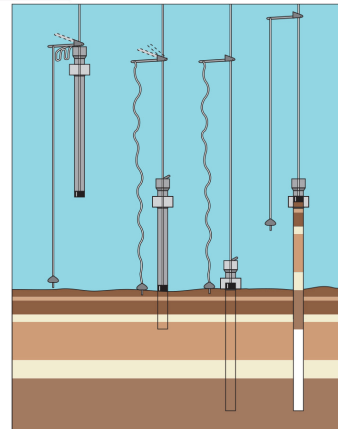
a

Clamshell Sampler



a

Piston Corer



b

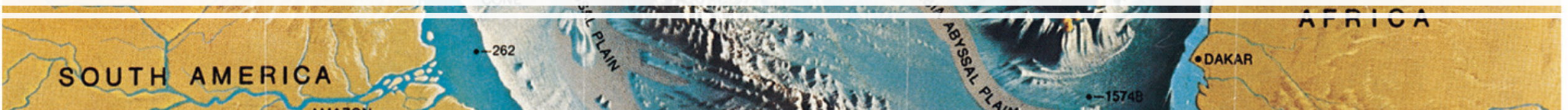
c

d

e



The submersed portions of the continental tectonic plates: **The Continental Margins**

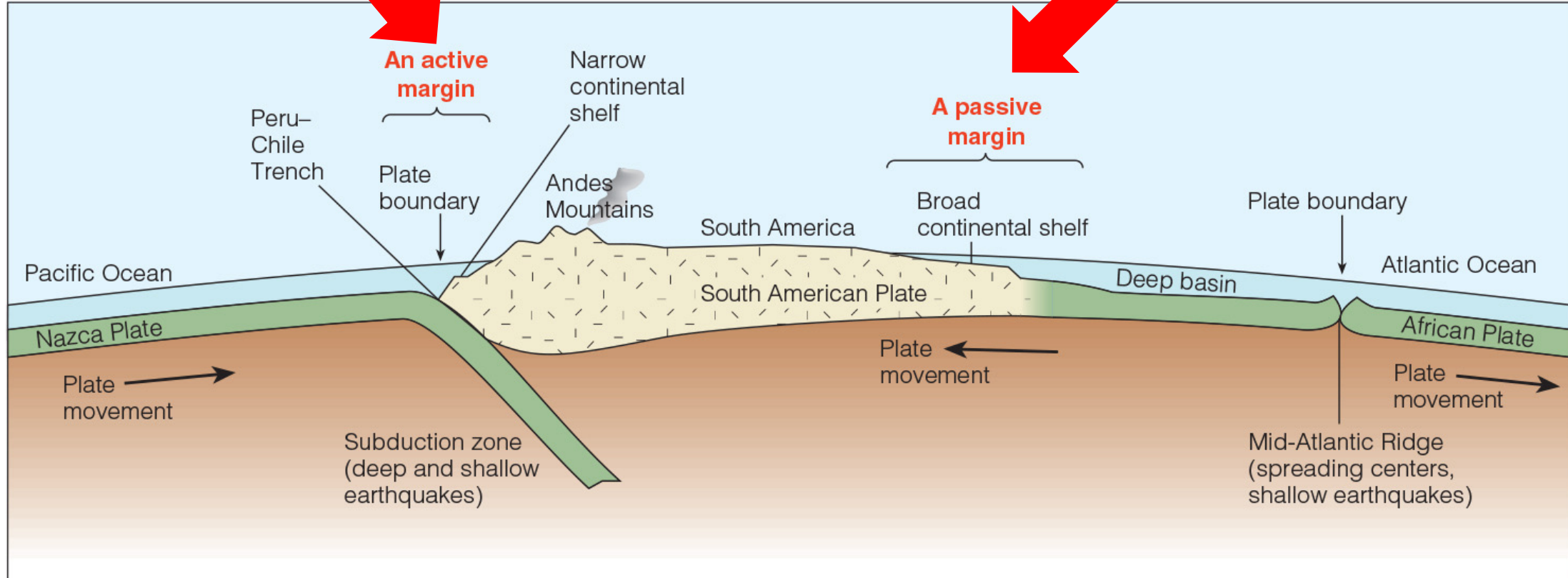


Active and Passive Continental Margins

Aka Pacific-type margins

Regardless the names, the continental margins surround all continents. Their type impacts their characteristics.

Aka Atlantic-type margins



- a** Typical continental margins bordering the tectonically *active* (Pacific-type) and tectonically *passive* (Atlantic-type) edges of a moving continent. (The vertical scale has been exaggerated.)

Figure 4.10 Active and passive margins.

Important areas of the continental margins:

Continental Shelves, **continental slope**, and **continental rise**

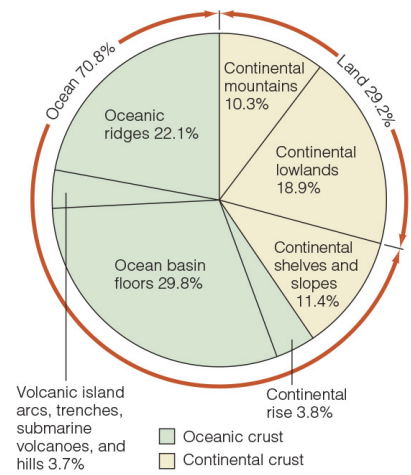
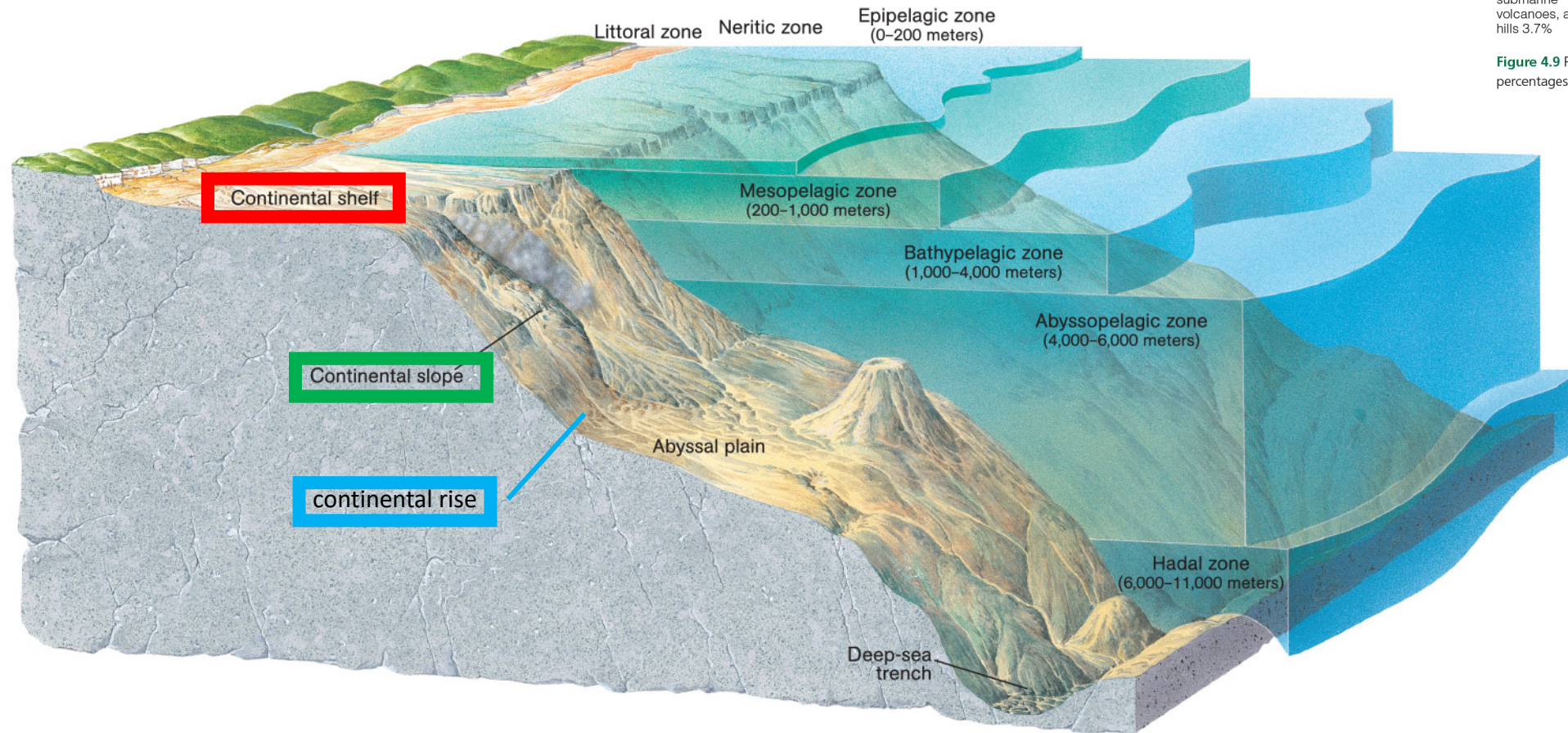
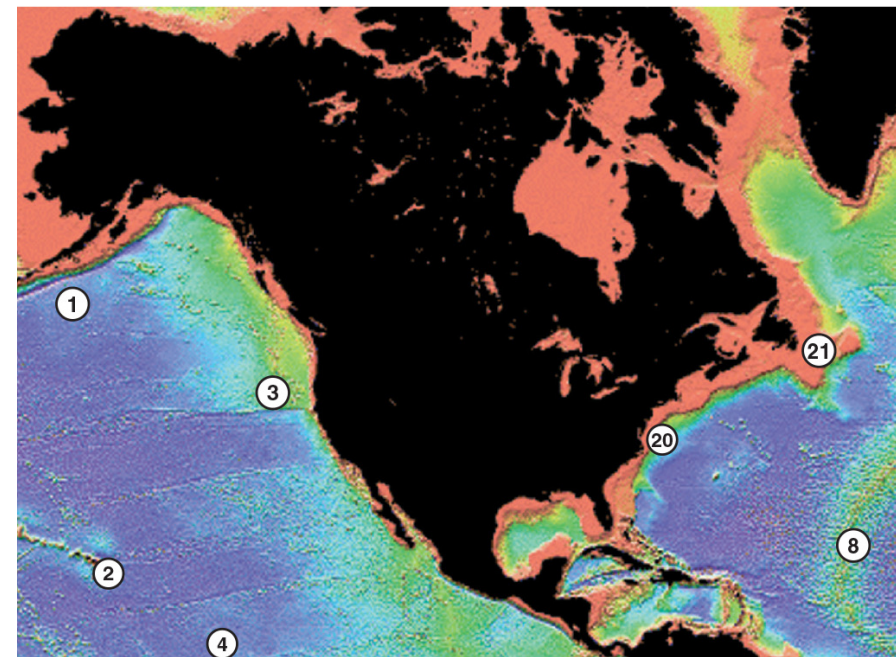


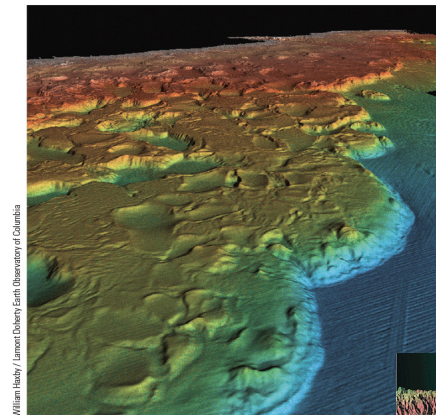
Figure 4.9 Features of Earth's solid surface shown as percentages of the planet's total surface.

Continental Shelves

- Covers only 7.4% of the Earth's ocean area, however, most of the resources we extract from the oceans come from areas over the continental shelves;
- Water column global average ~75 m.
- Continental shelves are characterized by gentle bottom slopes of 0.1° . For every kilometer away from the shore, the continental shelf gets up to 2 meters deeper (or ~9 ft per mile);
- **Continental shelf from a Passive (Atlantic-type) Continental Margin:** Formed from the thinning of the continental crust during the rift phase. They are usually wide (few hundreds of km) and shallow (up to 200 m). They are mainly covered by **sediments (erosion of continental rocks)** brought by rivers.
- **Continental shelf from an Active (Pacific-type) Continental Margin:** These are not as broad and flat than the passive type. Active-margin shelves have more varying topography, and its character may be determined more by plate movements (faulting, volcanism, and tectonic deformation) than by sedimentation.

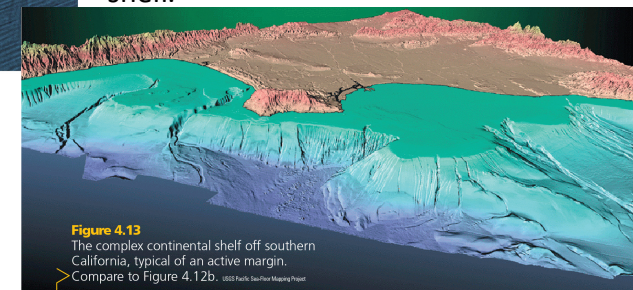


North American continental shelves are highlighted in red/orange colors on this map



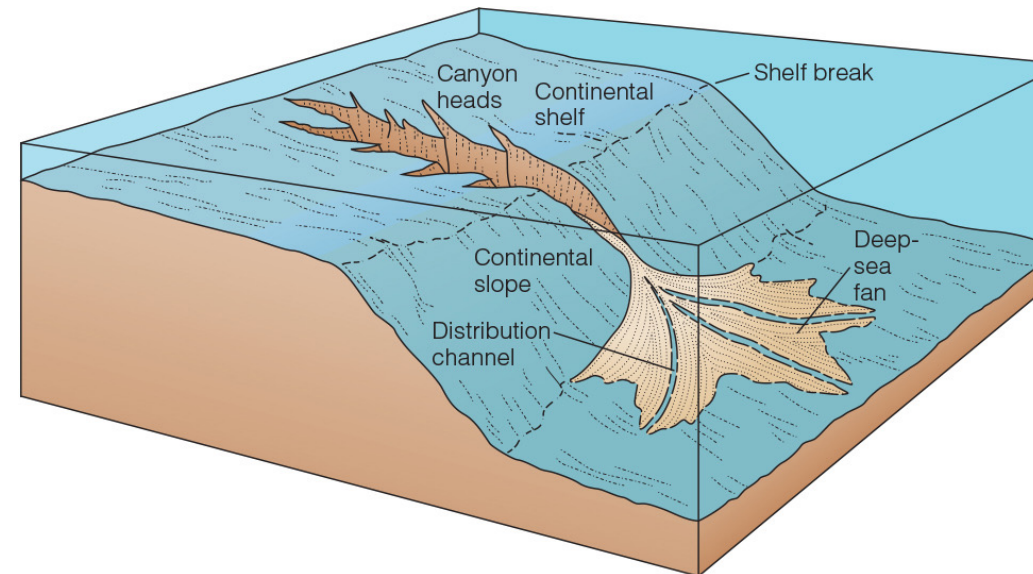
Passive Margin: Continental shelf off the coast of Louisiana mainly covered by sediments brought by the Mississippi River

Active Margin: California continental shelf.



Continental slope

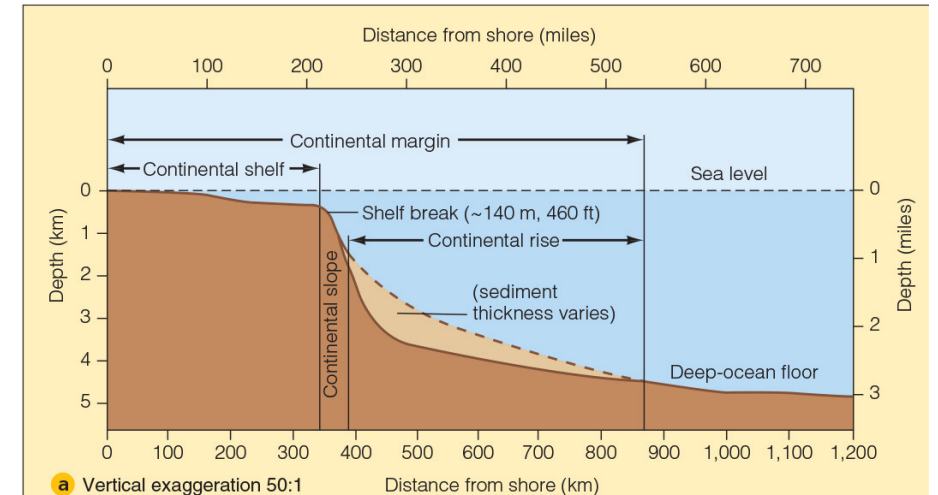
- Transition between the gently descending shelf and the deep-ocean floor (limit between continental and ocean crust).
- Formed by sediments that reaches the edge of the shelf (aka shelf break) and transported over the side.
- Active-margins slopes might also include sediments scraped off the subducted tectonic plate;
- They average on 20 km wide (12 miles) and have inclinations of 4° (70 m per km or 370 feet per mile). Therefore, their varying depth ranges from a few hundreds of meters to a few thousands.



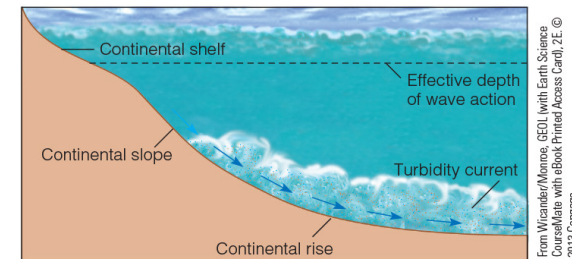
Continental Rise

On Passive Margins, the oceanic crust at the base of the continental slope is covered by a “apron” of accumulated sediments called the **Continental Rise**.

Its thickness and width are highly variable and its slope is gradual (something in between the slope and the shelf). **Due to its local-specific characteristics it is hard to attribute typical numbers.**



Sediments from the shelf descend the slope mostly via turbidity currents (“underwater avalanches”) and accumulate at the base of the slope



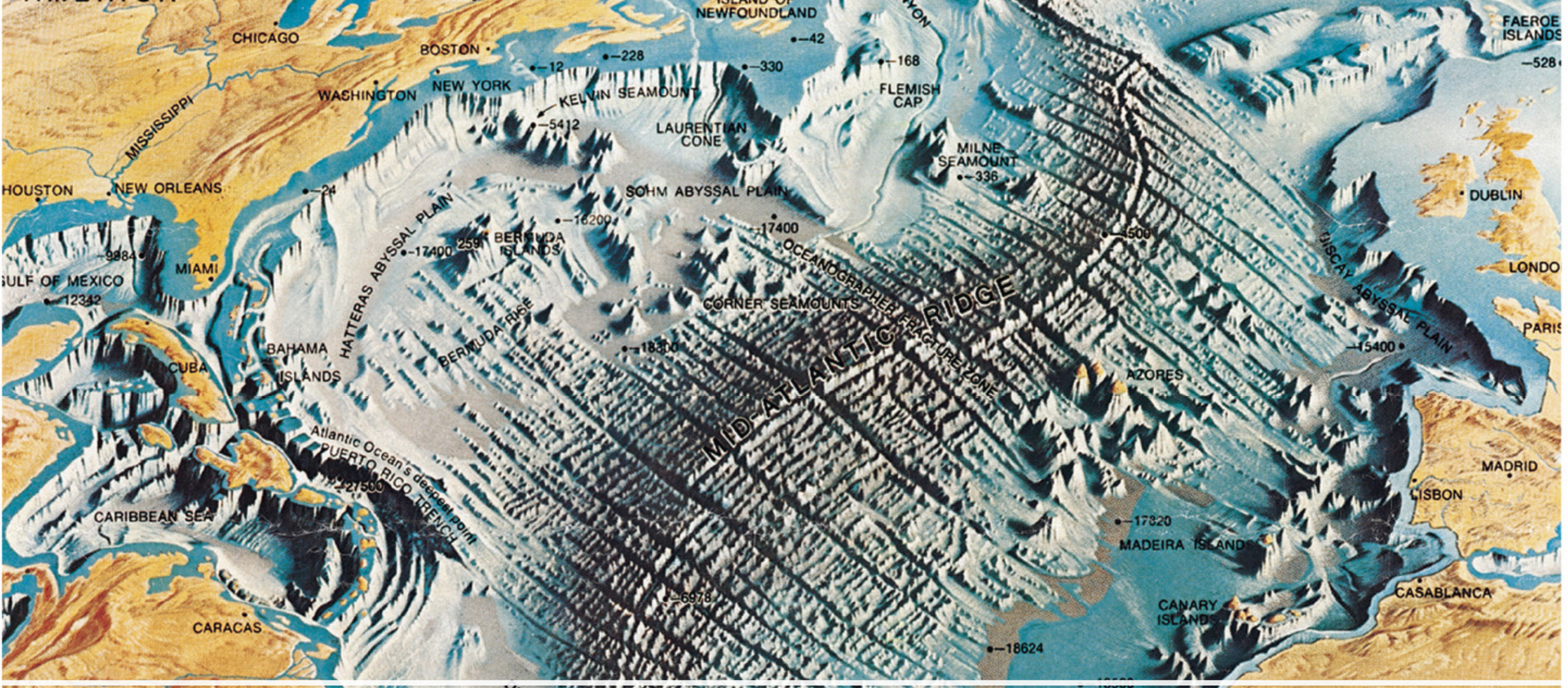
a A turbidity current is propelled not by the water within it but by gravity. Sediment instability causes the material to cascade down the continental slope and onto the continental rise.



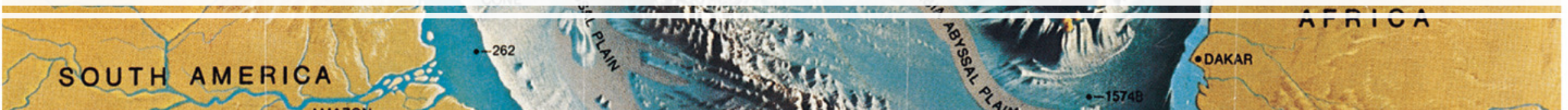
b A turbidity current flowing down a submerged slope off the island of Jamaica. The propeller of a submarine caused the turbidity current by disturbing sediment along the slope.

Question: Why do you think Active Margins don't present Continental Rises?

Answer: Any descended sediments would accumulate at the bottom of trenches and eventually subducted to the Mantle.



Abyssal plain and deep ocean features



Deep-ocean features cover ~60% of the Earth's solid surface. Most of it consists of Oceanic Ridges (~22%) and sediment covered plains (~30%, sediments thickness > 5 km).

Flat deep plains may be rimmed by trenches, interrupted by islands and volcanoes (active and extinct).

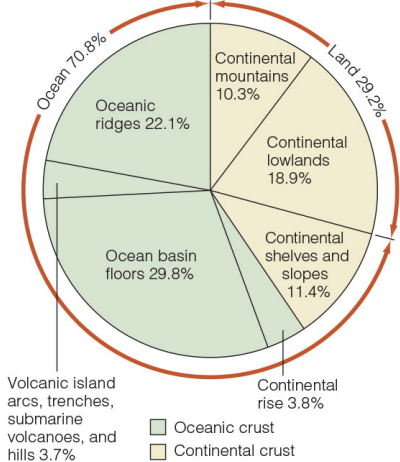
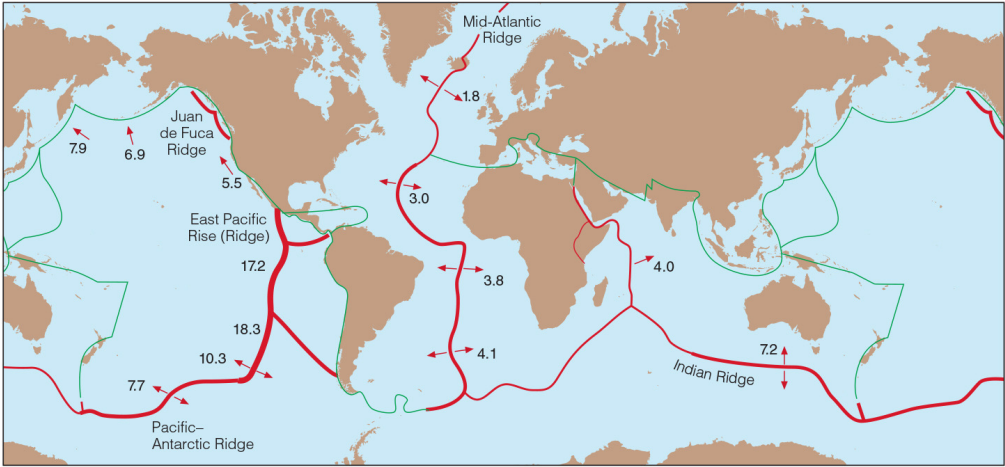
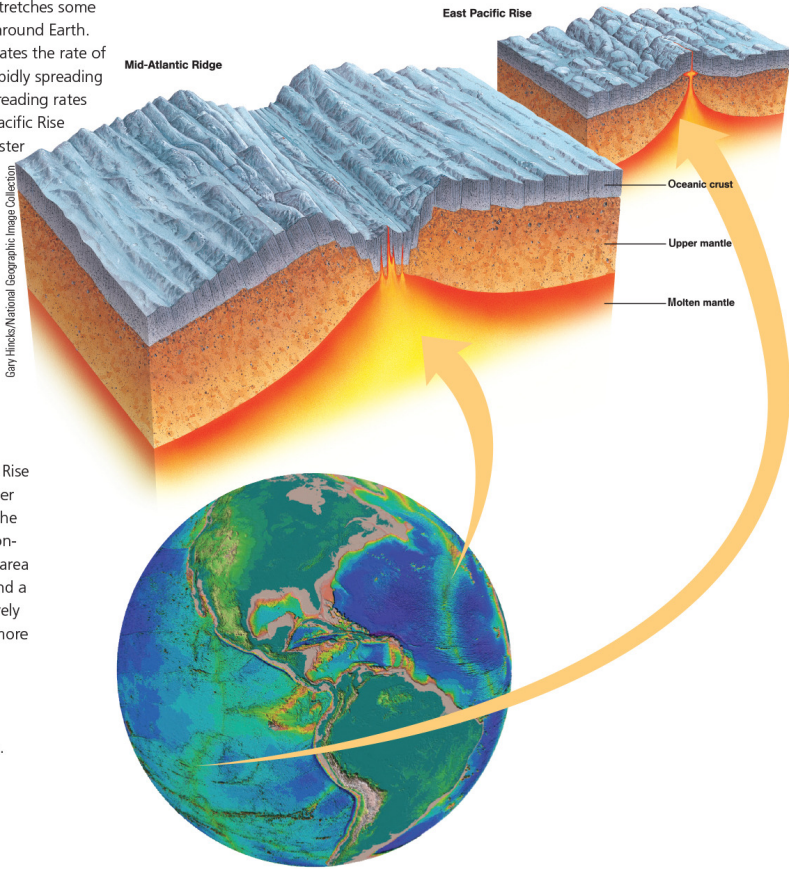


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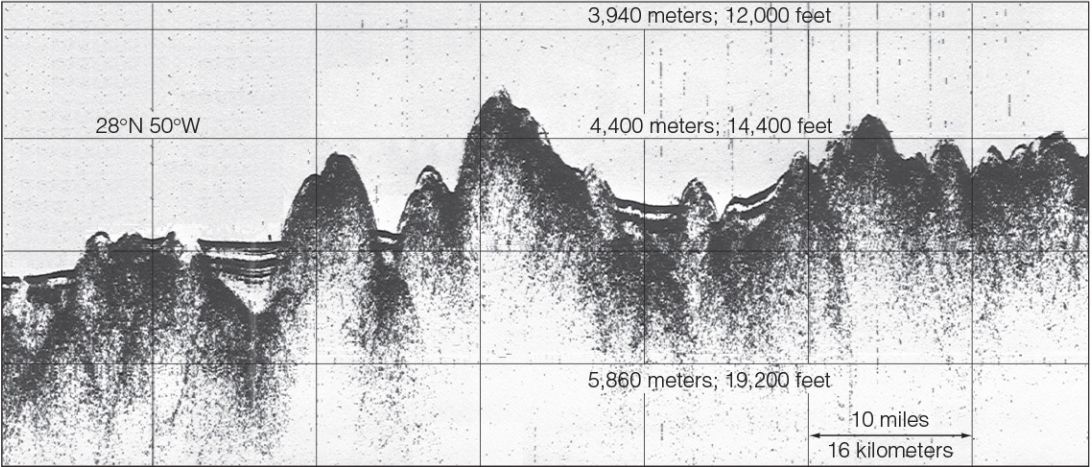
a The oceanic ridge system (in red) stretches some 65,000 kilometers (40,000 miles) around Earth. The thickness of the red lines indicates the rate of spreading for some of the most rapidly spreading sections, and the numbers give spreading rates in centimeters per year. The East Pacific Rise typically spreads about six times faster than the Mid-Atlantic Ridge. The thin green lines represent subduction zones.



b Rapid spreading at the East Pacific Rise spreads ridge features over a greater area. The slower spreading along the Mid-Atlantic Ridge (upper right) concentrates the features in a smaller area with a more vertical topography and a pronounced central rift. The relatively slow-spreading ridge is shown in more detail in Figure 4.20a.

Figure 4.18 The oceanic ridge system.

The Oceanic Ridges: Edge of divergent plates; intense volcanic activity; mountains that are thousands of meters tall.



Seismic profilers works in “a similar way” of echo sounders, but shock waves are used instead of sound.

b As this seismic profile shows, the rugged relief of the North Atlantic's oceanic ridge is gradually being buried by slowly accumulating sediments.

The Abyssal Plains: Their flatness come from vertically deposited and stacked thick layers of sediments. Although they are vast flat areas, all sorts of features are present.

Map of important topographic features

